



Simulation of weed emergence and leaf development in rice fields: An integrated machine learning and phyllochron approach

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ABSTRACT

Barnyardgrass (*Echinochloa crus-galli*) poses a significant challenge in rice production due to its aggressive emergence and morphological similarity to rice, complicating timely weed control. This study aimed to develop a robust, data-driven framework to accurately predict the emergence and leaf development of barnyardgrass across different rice establishment systems, with the ultimate goal of improving herbicide timing and control efficacy. Using 9 years of trial data, we constructed an integrated modeling framework combining a machine learning-based emergence model with a process-based phyllochron leaf development model. The framework was evaluated across three rice establishment systems: transplanting, dry direct seeding, and wet direct seeding. The ensemble emergence model achieved R^2 values of 0.92 (training) and 0.86 (testing), with mean absolute errors (MAE) of 5–6 % and 7–8 %, respectively. The phyllochron model yielded R^2 values of 0.64 (training) and 0.75 (testing). Independent validation trials conducted in 2024 showed that the differences between predicted and observed leaf emergence dates (Δ days) were narrowly distributed around zero, confirming the high temporal precision of the model. The integrated modeling framework demonstrated strong predictive accuracy for both emergence and leaf development of barnyardgrass. This tool offers practical utility for optimizing the timing and efficacy of herbicide applications across diverse rice production systems, contributing to more precise and sustainable weed management strategies.

1. Introduction

Weed emergence timing is a critical factor in crop–weed competition, particularly for rice production systems where early weed growth can severely impact yield. Barnyardgrass (*Echinochloa crus-galli*) poses a significant challenge to rice production due to its aggressive growth, high seed production (Zhang et al., 2017), and striking morphological similarities to rice, which enable it to effectively compete for limited resources (Awan et al., 2016; Chauhan and Johnson, 2011; Kraehmer et al., 2016). Despite the species having developed resistance to numerous widely used herbicides, which complicates control efforts, chemical weed control remains an efficient strategy for managing this troublesome weed (Rao et al., 2007; Talbert and Burgos, 2007). However, most herbicides were introduced years ago and face growing resistance or tighter restrictions, making their efficient use essential for sustainable management of barnyardgrass. Applying the right product at

the right rate and time is key (Krausz et al., 2000), yet timing is often overlooked. Notably, differences in rice establishment methods (such as direct seeding versus transplanting or water-seeding) create distinct environmental conditions that strongly influence barnyardgrass emergence (Yang et al., 2025). Dry direct-seeded rice systems, which lack continuous early flooding, often provide an ideal habitat for weeds, particular barnyardgrass, resulting in significant infestations. In contrast, flooded or transplanted rice systems suppress barnyardgrass germination and early growth, historically shifting weed populations toward more flood-tolerant species, such as barnyardgrass (Chin, 2001). These observations underscore the need for predictive modeling tools that can account for management-specific emergence patterns of barnyardgrass in rice cropping.

Predictive modeling of weed emergence is considered an invaluable component of integrated weed management (IWM) strategies (Grundy, 2003; Knezevic et al., 2002). By forecasting when and how intensely

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weeds will emerge, farmers can better time control measures – for example, applying herbicides at the time when the largest cohort of weeds is most susceptible. Effective application timing, informed by emergence models, can maximize control efficacy and minimize crop yield losses while potentially reducing herbicide inputs (Carey and Kells, 1995). Weed emergence models also assist in predicting subsequent flushes of weeds, guiding follow-up management and reducing the risk of weeds escaping control and replenishing the seedbank (Colucci et al., 2024; Grundy, 2003). Such considerations are particularly important for barnyardgrass management, as its dormancy-driven staggered germination can result in multiple emergence flushes throughout a growing season (Swanton et al., 2000). If barnyardgrass emergence timing can be predicted with sufficient accuracy, management interventions—whether chemical, mechanical, or cultural—can be strategically scheduled to target barnyardgrass during its most susceptible stages, thereby help devise integrated weed management (IWM) programs (Y. Zhang et al., 2024).

Traditional weed emergence models—both empirical and mechanistic—have contributed significantly to understanding weed phenology, but many suffer from oversimplification and limited adaptability. Empirical models such as double-logistic and Gompertz functions have shown promise in capturing emergence timing under specific conditions. For instance, Picapietra and Acciaresi (2021) used a double-logistic model to define a pre-emergence herbicide window between 348–399 GDDs for controlling *Echinochloa colona* in Argentina, achieving up to 99 % control. Similarly, Reinhardt Piskackova et al. (2021) reported that 90 % of *Amaranthus palmeri* emerged before July in North Carolina, demonstrating the utility of such models in practical weed management. However, these approaches often treat emergence as a single-stage event and rarely incorporate critical biological processes like dormancy loss, germination, and pre-emergence growth (Grundy, 2003). Mechanistic models have attempted to address these limitations by explicitly modeling environmental drivers such as temperature and moisture, yet they too often overlook interactions with management practices like tillage and flooding, which can strongly influence seed burial, light exposure, and dormancy (Benech-Arnold et al., 2000; Goulart et al., 2020). Although enhancing these models with additional variables such as tillage depth, photoperiod, or seedbank dormancy status may improve fidelity (Gonzalez-Andujar et al., 2016), such improvements increase data requirements and computational complexity, reducing practicality. Recent reviews have called for integrating real-time sensing, soft computing, and environmental feedbacks (Lundy et al., 2014; Marschner et al., 2024; Yousefi et al., 2014). Still, few existing frameworks holistically capture the biological, environmental, and operational factors necessary for robust prediction, particularly in rice–weed systems where crop establishment method (e.g., transplanting vs. direct seeding) alters emergence dynamics. These gaps underscore the need for more integrated modeling strategies that blend data-driven and biologically grounded approaches.

Recent advances in data-driven modeling offer new avenues to refine weed emergence predictions (Chantre et al., 2018). Machine learning (ML) techniques, in particular, enable models to learn complex, non-linear relationships from multi-dimensional data (e.g. weather variables, soil conditions, management practices) without a priori simplifying assumptions (Zhao et al., 2024). Researchers have begun exploring ML for weed emergence forecasting, reporting that approaches like artificial neural networks can outperform traditional regression models when multiple factors are considered (Chantre et al., 2018). By allowing inputs such as soil conditions, field management, and seed bank traits to be factored in simultaneously, ML models can potentially address the multi-factorial nature of weed emergence that static thermal time models struggle with. However, purely data-driven models may lack biological interpretability (AlQuraishi and Sorger, 2021). A black-box ML model might predict when weeds emerge, but it does not inherently describe weed developmental stages that are important for management (e.g. leaf number for herbicide application

timing). This is where process-based modeling remains crucial. The phyllochron concept – the interval in thermal time between the appearance of successive leaves – is a well-established process model for plant development (Gramig and Stoltenberg, 2007). For weeds like barnyardgrass, a phyllochron-based submodel can estimate how quickly a seedling progresses through leaf stages once it has emerged, given temperature conditions. Integrating such a mechanistic component with ML predictions could yield a more powerful tool: the ML algorithms can handle the when and how many (predicting emergence timing and density under diverse conditions), while the phyllochron model handles the how fast (simulating post-emergence leaf development and growth stage progression). This integrated ML–phyllochron framework marries the strengths of data-driven prediction with those of biological process modeling, providing both accuracy and interpretability. To our knowledge, this approach is innovative within the field of weed science, as earlier studies have predominantly utilized either empirical models (including machine learning) or process-based models independently, rather than integrating both methodologies into a hybrid framework.

The objectives of this study are to (i) develop a machine learning model to predict barnyardgrass emergence under three rice establishment methods, (ii) integrate a phyllochron-based process model to simulate subsequent leaf development for each emerging cohort, and (iii) validate the combined framework across multiple sites to optimize herbicide timing. While this study focuses on barnyardgrass in rice systems, the modeling framework is broadly adaptable: by adjusting cohort thresholds and species-specific phyllochron parameters, it can be extended to other weed species and cropping systems, offering a generalizable approach for weed emergence and growth stage prediction in support of precision weed management.

2. Methods

2.1. Barnyardgrass emergence and leaf development

In rice production systems, the emergence dynamics of barnyardgrass are strongly influenced by the water regimes associated with different crop establishment methods (Fig. 1). Rice seedlings are pre-grown in nurseries and transplanted once they reach a uniform developmental stage. These established seedlings rapidly form a canopy and root system that suppress barnyardgrass emergence. In contrast, wet direct seeding (Fig. 1, middle panel) involves simultaneous sowing of rice and barnyardgrass seeds in continuously moist conditions, promoting rapid weed germination and potentially giving barnyardgrass a competitive edge in the absence of timely intervention. Dry direct seeding (Fig. 1, bottom panel) often leads to staggered or delayed rice emergence due to initial soil dryness, creating a window for barnyardgrass to establish more effectively relative to the crop.

Because water availability and crop establishment methods interact in complex, nonlinear ways, traditional emergence models often fail to capture the nuanced dynamics of barnyardgrass emergence. Machine learning offers a suitable alternative for predicting weed emergence under these variable conditions: it can learn from heterogeneous datasets, identify intricate relationships among multiple factors (e.g., temperature, soil moisture, day length, seed bank dynamics), and make predictions without relying on a complete mechanistic understanding of the emergence process. By contrast, leaf development for barnyardgrass follows a more direct and well-characterized mechanism—primarily driven by temperature—making it simpler to represent using a process-based model. This mechanistic clarity reduces the need for large, diverse datasets and lends itself to relatively straightforward parameterization, thereby complementing the machine learning approach used for the more complex task of predicting weed emergence.

2.2. Observation data acquisition

Field trials were conducted from 2023 to 2024 under three rice

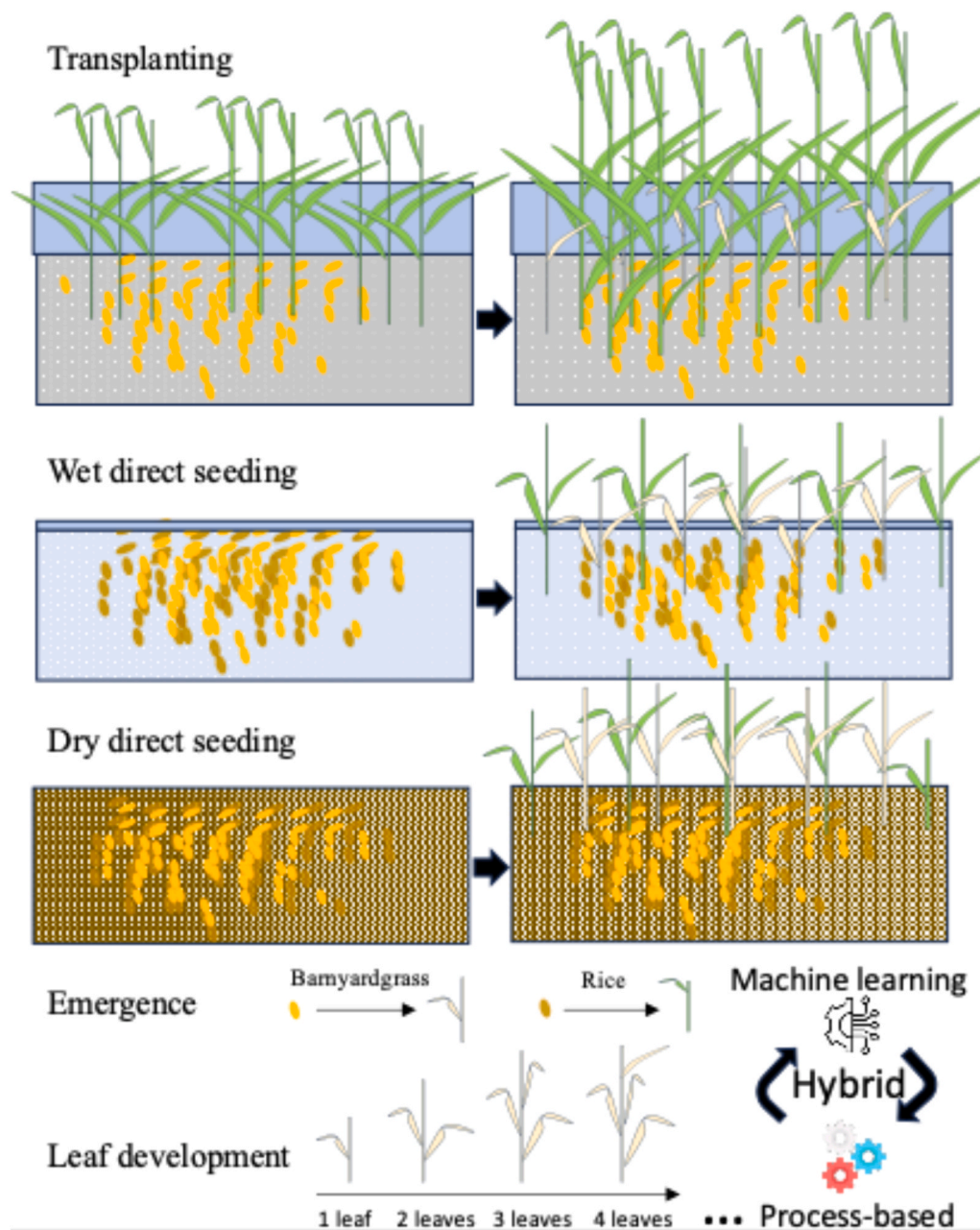


Fig. 1. Illustration of weed emergence and leaf development under three crop establishment systems.

establishment methods—transplanting, dry direct seeding, and wet direct seeding—to capture barnyardgrass (*Echinochloa crus-galli*) emergence patterns (Fig. 2). Each site was a normally managed field with a known weed management history. Frames were placed to capture spatial variability, thereby representing a range of weed seed bank conditions under otherwise uniform field management. To quantify emergence, 20 fixed 1×1 m frames were placed randomly within each trial area. Seedling counts in these frames were recorded twice weekly for the first two weeks, followed by biweekly observations, yielding a total of six emergence assessments per trial. Final weed counts were converted to percentages relative to each trial's maximum observed density for cross-site consistency. Aside from a single pre-trial burndown glyphosate application, all other agronomic inputs (e.g., crop variety, insecticides, fungicides) remained uniform. The wet direct seeding experiments were located near 34.86° N, 136.25° E (Mie Prefecture), the dry direct seeding trials near 31.53° N, 130.5° E (Kagoshima Prefecture), and the transplanting experiments spanned various climatic regions in

northern, central, and southern Japan (e.g., Hokkaido, Miyagi, Niigata, Ibaraki, Okayama, and Saga).

Barnyardgrass leaf development data were extracted in November 2023 from the database maintained by the Japan Association for Advancement of Phyto-Regulators (JAPR) (<https://japr.or.jp/>). Over the past 50 years, JAPR has partnered with research institutes and public organizations across Japan, generating extensive trial data on plant protection agents (including herbicides). For this study, records published between 2016 and 2023 were reviewed, focusing on weed-response variables such as percentage emergence and growth stage (expressed as leaf number).

The distribution of observational data across the three crop establishment systems reveals distinct patterns in both emergence and leaf development. As shown in Fig. 3 transplanting data tend to cluster around lower to midrange emergence percentages and a narrower range of calendar days, reflecting a more uniform planting schedule. Direct seeding, particularly under wet conditions, spans a wider set of

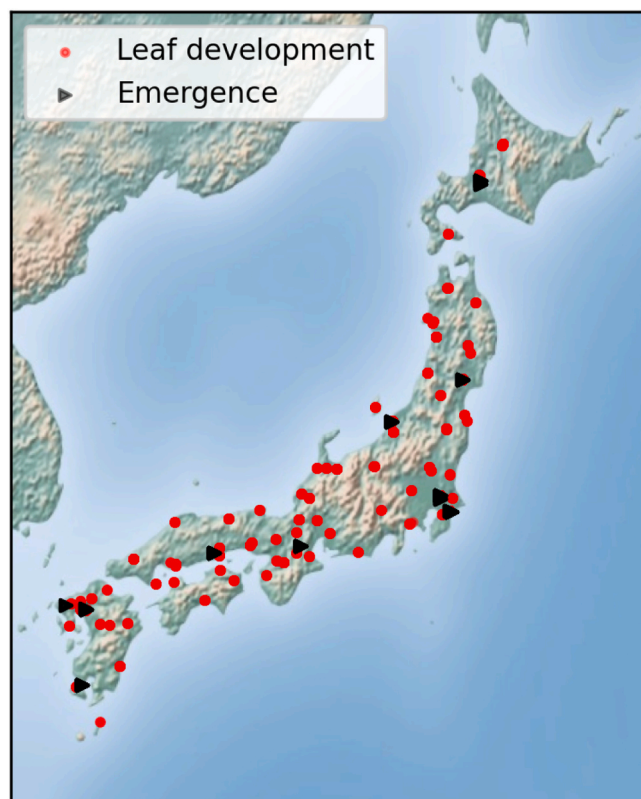


Fig. 2. The spatial distribution of seedling emergence and phenology data of *Echinochloa crus-galli* in rice fields in Japan.

emergence values—including higher percentages—and covers a broader window of establishment days. Leaf development stages likewise follow a narrower distribution in transplanting systems, whereas both wet and dry direct seeding account for a wider spread of leaf numbers. These differences underscore the diversity of agronomic practices and resulting weed populations across the dataset.

2.3. Weather data

We acquired weather data from DTN (<https://www.dtn.com/weather/>), a reputable commercial provider, which included air temperature, precipitation, and soil temperature measurements from the crop establishment date throughout the season. From the raw DTN weather data, we calculated key indices including air Growing Degree Days (GDD), based on daily mean air temperature with a 10–30°C threshold (Fig. 4); soil GDD, computed from soil temperature using the same thresholds (Fig. 4); and cumulative precipitation, measured from the date of crop establishment to the date of each observation. These indices provide a robust basis for our crop modeling by capturing the thermal time accumulation and water input dynamics that directly influence crop growth, canopy microclimate, and disease progression.

2.4. Model development

2.4.1. Emergence model training (ML)

A series of machine learning (ML) approaches was developed to predict barnyardgrass (*E. crus-galli*) emergence rates under three rice establishment systems: transplanting, dry direct seeding, and wet direct seeding. In each system, preprocessed data were divided into training (70 %) and testing (30 %) sets using a fixed random seed (random_state=42) to ensure reproducibility. Relevant predictor variables were selected based on the establishment method; for instance, transplanting models incorporated days after transplanting, latitude, longitude,

establish day-of-year (DOY), and environmental indices such as growing degree days (GDD), precipitation, and soil GDD. In dry and wet direct seeding, the focus shifted to variables like days after establishment, GDD, precipitation, soil GDD, and, for dry systems, soil moisture accumulation (Table 1).

Because barnyardgrass emergence depends on multiple, often interacting factors—including water management, tillage operations, weather conditions, and crop management—six ensemble-based ML regressors were chosen (Table 2). These included AdaBoostRegressor, BaggingRegressor, GradientBoostingRegressor, RandomForestRegressor, LGBMRegressor, and XGBRegressor. Ensemble methods were preferred for their capacity to capture the complex, nonlinear relationships inherent in weed emergence and their demonstrated robustness against overfitting, particularly with large, high-dimensional datasets.

Model training followed a rigorous protocol involving grid search with 5-fold cross-validation using negative mean squared error as the scoring metric (Fig. 5). Each algorithm underwent hyperparameter tuning according to predefined grids (Table 3 and STable 2). Parameters such as the number of estimators, maximum features, tree depth, learning rates, and sampling strategies were systematically varied to identify the best configuration. After the grid search, the optimal model from each algorithm was retrained on the full training set and evaluated on the test set using the coefficient of determination (R^2) and mean absolute error (MAE). Models that provided feature importance metrics were further examined to determine each predictor's contribution to overall emergence rate predictions. This integrated framework ensures accurate modeling by combining key predictors, optimized ensemble regressors, and rigorous hyperparameter tuning.

2.4.2. Phyllochron leaf development model

For our leaf development model after emergence, we employ a phyllochron approach that simulates the thermal time requirements for successive leaf appearance. In this model, a base temperature of 10°C and a ceiling temperature of 30°C are applied (Zhang et al., 2023). Any daily mean air temperature below 10°C contributes 0 GDD (Growing Degree Days), while temperatures above 30°C are capped, resulting in a maximum daily GDD of 20°C. The phyllochron, defined as the thermal time interval required for the appearance of each new leaf, is derived from a linear regression between observed barnyardgrass leaf numbers and the cumulative GDD from crop establishment. To simulate leaf numbers at a 0.5 scale (i.e., accounting for fractional leaf emergence such as BBCH 11.5), we applied linear interpolation using the fitted regression equation ($GDD = 37.10 \times \text{leaf number} + 3.38$). To simulate leaf numbers at a 0.5 scale (i.e., accounting for fractional leaf emergence such as 1.5 leaves), we applied linear interpolation based on the fitted linear regression model relating leaf number to accumulated GDD. Specifically, the GDD requirement for a half-leaf increment was estimated by adding half the per-leaf GDD increment predicted by the model to the GDD of the preceding full-leaf stage.

2.4.3. Cohort aggregation & integration

The integration of the emergence model with the leaf development model is achieved through a multi-step process that groups the weed population into discrete cohorts and then simulates leaf development for each cohort individually before combining the results (Fig. 6). First, the emergence model processes environmental data (such as soil temperature, air temperature, precipitation, and other key parameters) to predict a cumulative emergence rate over time using the machine learning approach with the best performance. Based on the predicted emergence percentages, the model groups the weeds into 10 cohorts or flushes, each representing a 10 % increment of cumulative emergence (e.g., 0–10 %, 10–20 %, etc.).

Once a cohort reaches its defined emergence percentage, the leaf development model is activated for that cohort. After simulating the leaf development for each of the 10 cohorts, the outputs—each reflecting the specific emergence dynamics and leaf development of that cohort—are

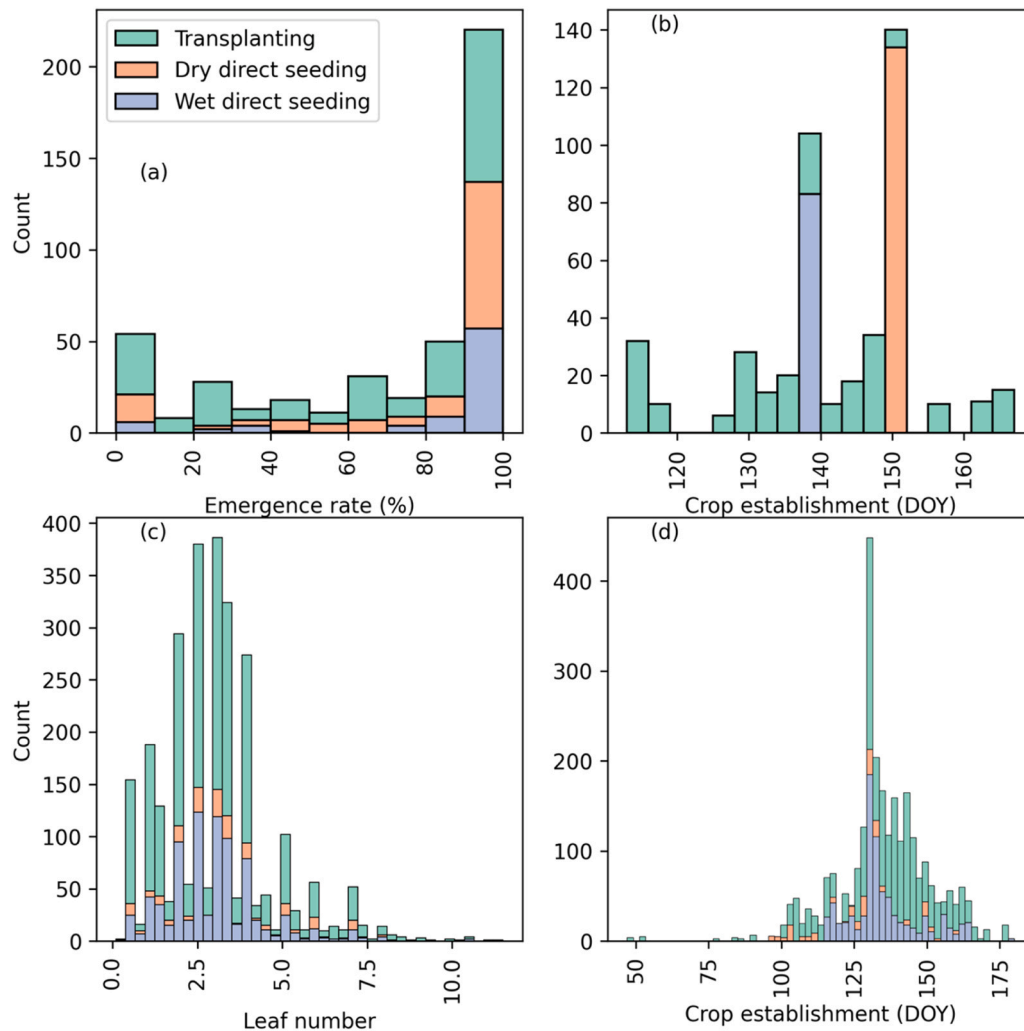


Fig. 3. Trial data counts and distributions across crop establishment systems. (a) Emergence rates distribution for transplanting, dry direct seeding, and wet direct seeding. (b) Distribution of crop establishment dates corresponding to emergence observations. (c) Leaf number distribution across the three crop establishment methods. (d) Distribution of crop establishment dates for leaf number observations. DOY: Day of Year.

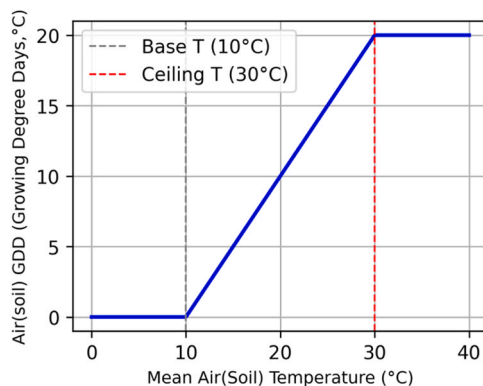


Fig. 4. Illustration of converting mean air and soil temperature to Growing Degree Days (GDD).

aggregated to form a comprehensive profile of the overall weed or crop canopy development. This integrated framework allows for a more accurate representation of how heterogeneous emergence patterns contribute to final canopy structure and function, and it provides valuable insights for optimizing crop management strategies.

Table 1

The selected predictor variables used for training the emergence rate prediction models for each crop establishment system.

Establishment System	Key Predictor Variables	Units
Transplanting	Days after crop establishment	days
	GDD from crop establishment	°C·day
	Precipitation from crop establishment	mm
	Soil GDD from crop establishment	°C·day
Dry Direct Seeding	GDD from crop establishment	°C·day
	Precipitation from crop establishment	mm
	Soil GDD from crop establishment	°C·day
	Soil moisture accumulation from crop establishment	%·day
Wet Direct Seeding	Days after establishment	days
	GDD from crop establishment	°C·day
	Precipitation from crop establishment	mm
	Soil GDD from crop establishment	°C·day

2.5. Model evaluation

In evaluating the predictive performance of the six machine learning regression models (i.e., AdaBoostRegressor, BaggingRegressor, GradientBoostingRegressor, RandomForestRegressor, LGBMRegressor, and XGBRegressor) for barnyardgrass emergence rates, two key metrics were

Table 2

Summary of machine learning regression models used for predicting barnyard-grass emergence rates.

Model	Description	Reason for Selection
AdaBoostRegressor	Combines multiple weak learners to boost predictive accuracy (Freund and Schapire, 1997; Schapire, 2013).	Its ensemble structure captures complex, nonlinear interactions and is robust against overfitting.
BaggingRegressor	Trains several instances of a model on bootstrapped subsets and aggregates their predictions (Aslam et al., 2023).	Reduces variance and enhances stability, making it well-suited for high-dimensional datasets.
GradientBoostingRegressor	Constructs an additive model in a forward stage-wise manner to optimize a differentiable loss function (Iaccarino et al., 2024).	Offers high predictive power and effectively handles intricate interactions within complex datasets.
RandomForestRegressor	Builds an ensemble of decision trees and outputs the average prediction from each tree (Graw et al., 2021).	Excels with large, high-dimensional datasets and provides robust predictions through variance reduction.
LGBMRegressor	Based on the LightGBM framework, this efficient and scalable algorithm is designed for speed and high accuracy (Chand and Divyab, 2022).	Delivers rapid and precise predictions, efficiently managing large datasets with numerous features.
XGBRegressor	Developed under the XGBoost framework, this flexible algorithm is powerful and scalable (Mariadass et al., 2022).	Renowned for its performance and scalability, it adeptly captures complex interactions in large datasets.

employed: the Mean Absolute Error (MAE, s.eq.1), Root Mean Squared Error (RMSE, s.eq.2), and the coefficient of determination (R^2). These metrics allowed us to quantify how well each model captures the complex, nonlinear interactions among the predictor variables influencing emergence.

The coefficient of determination (r^2) is given by:

$$r^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where $\bar{y}$ is the mean of the observed emergence rates. The r^2 value reflects the proportion of variance in the observed data that is explained by the model. An r^2 of 1 denotes a perfect fit, while an r^2 of 0 indicates that the model does not improve upon simply using the mean of the observed values as a predictor.

Using the integrated emergence and leaf development model, we assess its temporal accuracy by comparing the predicted and observed dates for a specific leaf number. This evaluation is based on independent trials conducted in 2024 that were not used during model training. The day differences directly measure the model's precision in forecasting leaf appearance timing, with smaller differences indicating higher accuracy.

3. Results

3.1. Performance of machine model for emergence rate prediction

For emergence prediction, RandomForest, XGB, and AdaBoosting show similarly strong performance, with R^2 values around 0.90–0.92 on the training set and approximately 0.84–0.86 on the test set (Fig. 7a).

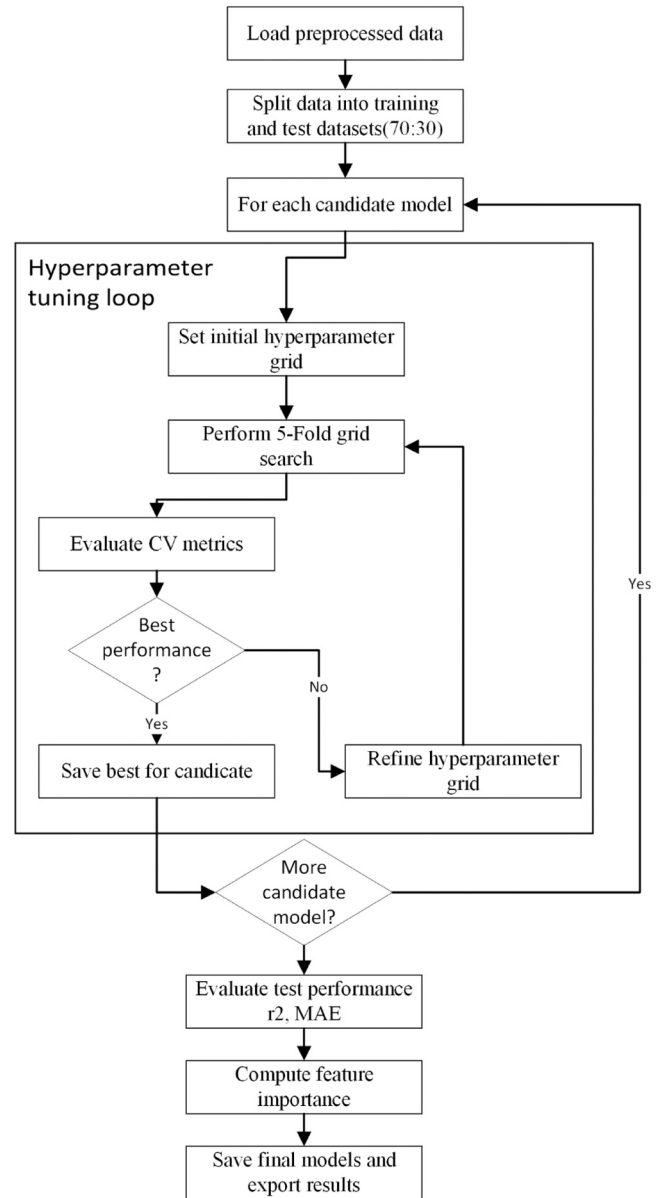


Fig. 5. Machine learning model training workflow for emergence rate prediction models.

Table 3

Optimized hyperparameters for each machine learning model.

Model	Optimized Hyperparameters
AdaBoostRegressor	n_estimators = 30; learning_rate = 0.1; loss = exponential
BaggingRegressor	n_estimators = 20; max_features = 2; max_samples = 3
GradientBoostingRegressor	n_estimators = 200; learning_rate = 0.1; max_depth = 4; min_samples_split = 5; min_samples_leaf = 15
RandomForestRegressor	n_estimators = 30; max_features = 4; max_depth = 20; min_samples_split = 5; min_samples_leaf = 10
XGBRegressor	n_estimators = 200; learning_rate = 0.1; max_depth = 4; subsample = 0.8; colsample_bytree = 0.8
LGBMRegressor	n_estimators = 200; learning_rate = 0.1; max_depth = 4; subsample = 0.8; colsample_bytree = 0.8

GradientBoosting follows closely (≈ 0.86 training, 0.82 test), while Bagging and LGBM perform worse, with R^2 values of about 0.73 / 0.67 and 0.70 / 0.60, respectively. Error-based metrics support these

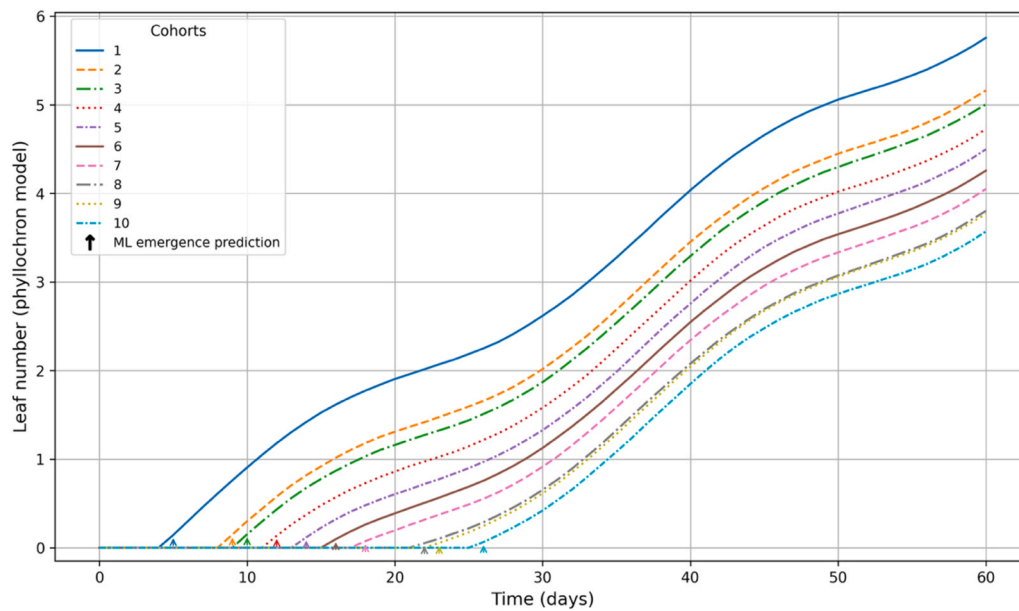


Fig. 6. Integrated simulation framework for weed development illustrating the hybrid modeling approach. The machine learning (ML) model predicts cumulative emergence rates, grouping the population into 10 cohorts, each representing 10 % increment). For each cohort, the phyllchron-based model simulates leaf development, employing linear interpolation to capture fractional leaf emergence. Finally, the outputs from all cohorts are aggregated to generate an overall canopy development profile.

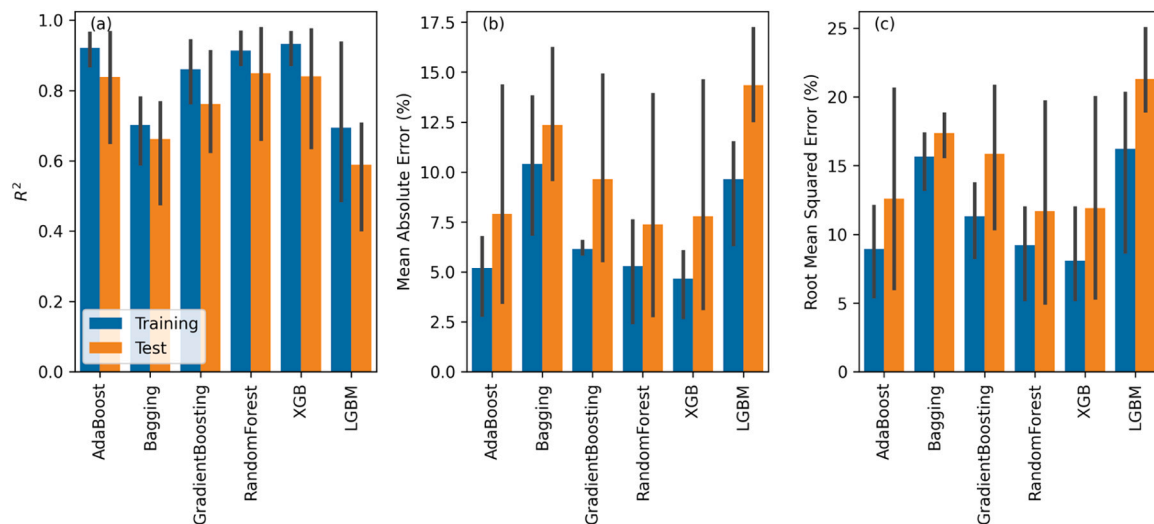


Fig. 7. The performance of six machine learning regression models was evaluated in predicting the emergence of *Echinochloa crus-galli* in rice fields under three crop establishment methods—transplanting, dry direct seeding, and wet direct seeding—using (a) the coefficient of determination (R^2), (b) mean absolute error (MAE), and (c) root mean squared error (RMSE). The vertical lines indicate the standard deviation. The coefficient of determination is defined as, where is the residual sum of squares ($(y_{\text{true}} - y_{\text{pred}})^2$).sum() and is the total sum of squares ($(y_{\text{true}} - y_{\text{true.mean}})^2$).sum(). The best possible score is 1.0 and it can be negative (because the model can be arbitrarily worse). A constant model that always predicts the expected value of y , disregarding the input features, would get a score of 0.0.

rankings. RandomForest, XGB, and AdaBoosting all achieve low Mean Absolute Error (MAE)—roughly 5–6 % in training and 7–8 % in testing (Fig. 7b). GradientBoosting shows moderately higher MAE (≈ 6 % training, 10 % test), while Bagging and LGBM exceed 10 % in training and 12 % in testing. The same trend holds for Root Mean Squared Error (RMSE) (Fc): RandomForest, XGB, and AdaBoosting exhibit the lowest RMSE (≈ 9 –10 % training, ≈ 12 % test), GradientBoosting is slightly higher (≈ 12 % training, 15 % test), and Bagging and LGBM again show the largest errors (≥ 15 % training, ≥ 18 % test). These results demonstrate that RandomForest, XGB, and AdaBoosting have similarly strong generalization performance, outperforming Bagging and LGBM, which show consistently larger prediction errors.

Key determinants of *barnyardgrass* emergence across different rice establishment methods

The random forest model showed that soil GDD accumulated from crop establishment (CE) emerged as the most influential feature for transplanting (blue bar), surpassing all other variables in explaining barnyardgrass (*E. crus-galli*) emergence (Fig. 8). For dry direct seeding (orange bar) and wet direct seeding (green bar), GDD accumulated from CE and days after CE also exhibited considerable importance, albeit at lower magnitudes compared to soil GDD in the transplanting scenario. Meanwhile, precipitation accumulated from CE and soil moisture accumulated from CE showed moderate to lower importance across all three establishment methods. Overall, these results highlight the pivotal

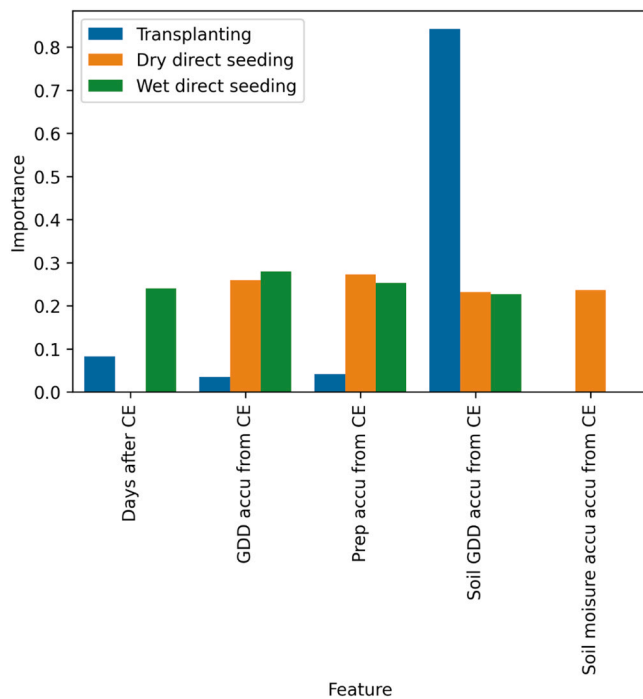


Fig. 8. Feature importance identified by the random forest model with best performance for predicting the emergence of *Echinochloa crus-galli* across three rice crop establishment methods: transplanting, dry direct seeding, and wet direct seeding. CE: Crop establishment.

role of soil-based thermal time in driving *Barnyardgrass* emergence under transplanting, whereas GDD (air-based) and temporal factors (days after CE) appear to exert relatively greater influence in direct-seeded systems.

3.2. Relationship between leaf development and GDD

Fig. 9 illustrates the linear relationship between leaf number and the accumulated growing degree days (GDD) from crop establishment, with a regression line of $y = 37.1x + 3.38$. The training data yield an r^2 of 0.64, while the test data achieve a higher r^2 of 0.75, indicating that the model captures leaf development with moderate to high accuracy. Fig. 9

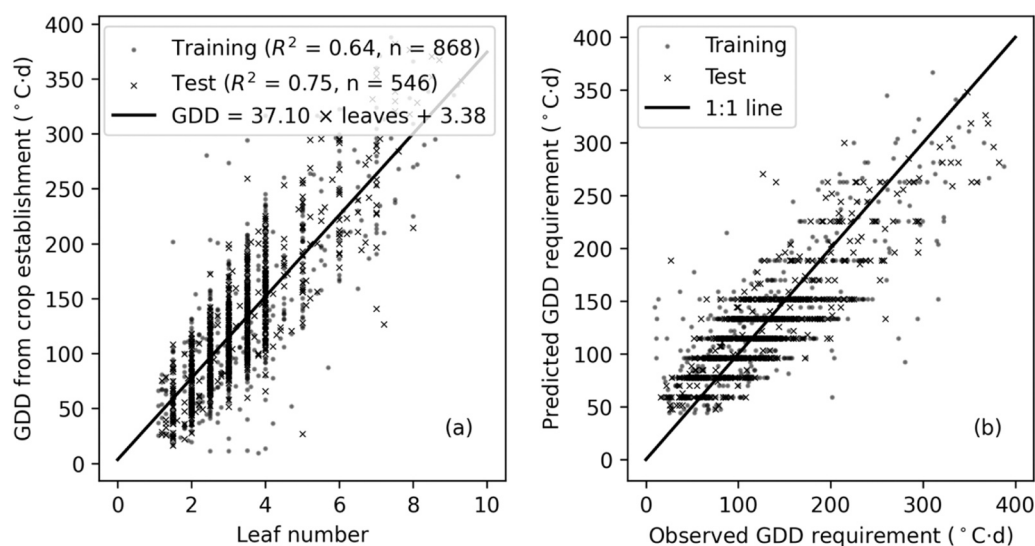


Fig. 9. Linear regression model to estimate growth degree days (GDD) requirements for leaf development of *Echinochloa crus-galli*. (a) is the linear regression model performance on train and testing dataset. (b) is the predicted GDD requirements versus observed GDD requirements for different leaf development stages.

(b) compares the observed and predicted GDD requirements on a 1:1 reference line. Here, the distribution of points around the line suggests that the model generally approximates the thermal time needed for leaf appearance. The training set clusters closely to the line, and the test set maintains a similar trend, underscoring the model's capacity to generalize. Overall, these results confirm that the model effectively links leaf number to GDD accumulation and reliably estimates the GDD requirement for leaf development, even when tested on unseen data. A residual-versus-predicted plot (Fig. S1) also shows no clear heteroscedastic pattern, indicating that model errors are approximately homoscedastic across the predicted GDD range.

3.3. Simulated versus observed emergence and leaf development

The simulated emergence and leaf development patterns of *Barnyardgrass* across six trial locations under three crop establishment methods (dry direct seeding, wet direct seeding, and transplanting) (Fig. 10). Compare to the observed emergence rate over time, the model captures the progression of *Barnyardgrass* leaf development at different latitudes and management methods, as evidenced by a general alignment between the predicted contours and the observed data. Nevertheless, some discrepancies appear, particularly at later leaf stages or higher emergence percentages, suggesting that localized environmental factors or field practices may influence weed growth. Despite these variations, the results highlight the model's ability to integrate varying environmental conditions and establishment strategies, offering a coherent representation of *Barnyardgrass* emergence and leaf dynamics across diverse locations.

3.4. Distribution of prediction error in leaf development by the integrated model

Fig. 11 shows the frequency distribution of delta days, which measures the difference in days between predicted and observed leaf emergence events. The majority of delta days values cluster near zero, indicating that the integrated random forest and phyllochron model accurately predicts leaf development timing for most observations. A smaller group of points with negative delta days (roughly -10 to -5) suggests that the model occasionally predicts emergence earlier than observed, whereas the cluster in the 5–10 range indicates some later predictions. Overall, the narrow dispersion around zero underscores the model's effectiveness in capturing the temporal dynamics of leaf

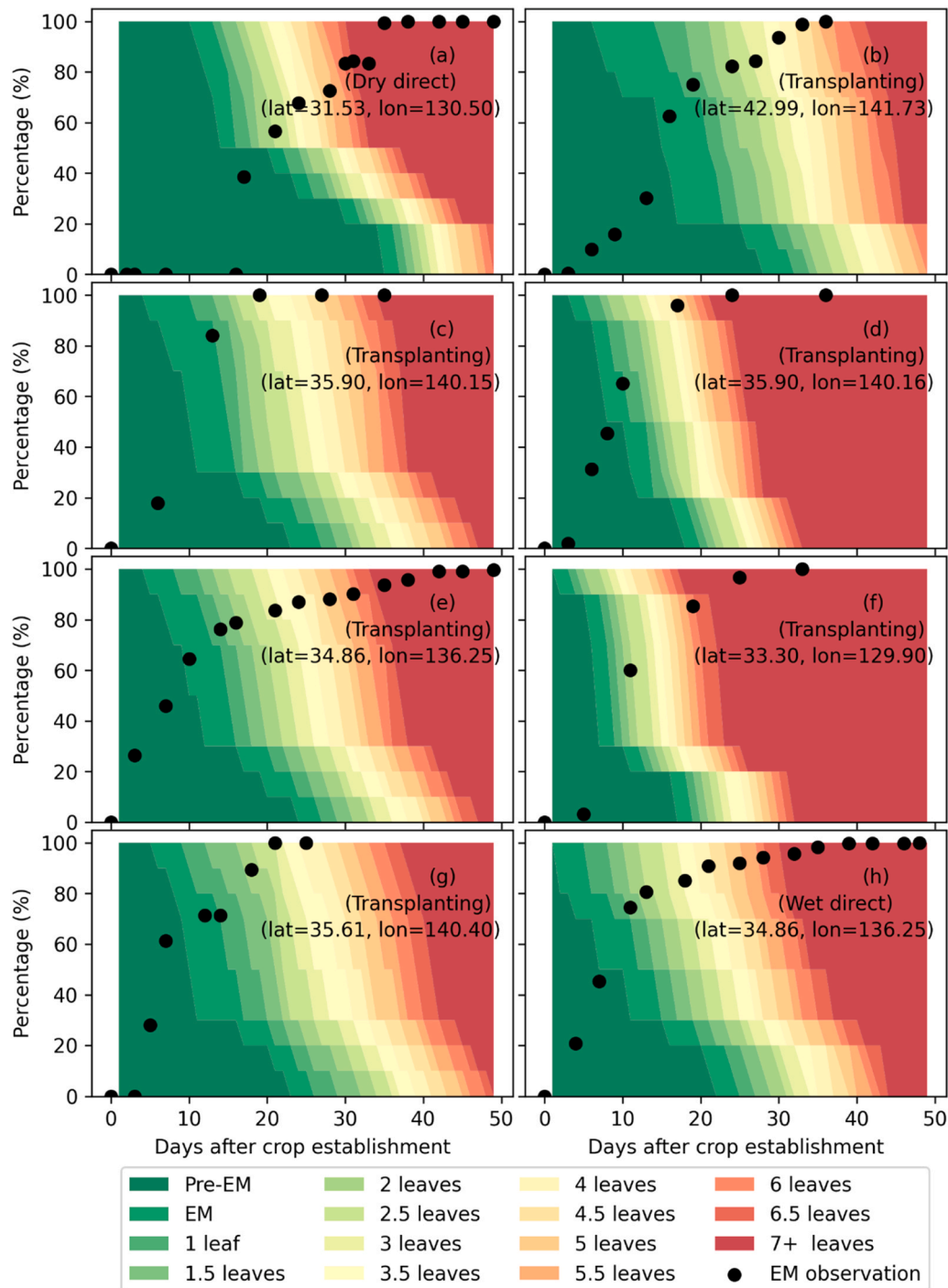


Fig. 10. Simulation results for the weed emergence rate and leaf development, along with emergence rate observations, for Barnyardgrass across eight trial locations employing three crop establishment methods. EM: Emergence; Lat: Latitude; Lon: Longitude. Black points represent the observed emergence rates.

appearance, while the outlier groups highlight potential site- or management-specific deviations.

4. Discussion

In this study, we present the first integration of a machine learning-based emergence model with a phyllochron-driven leaf development model to simulate barnyardgrass dynamics in rice systems. Leveraging multi-year, multi-site data across transplanting, dry direct seeding, and wet direct seeding methods, the emergence model effectively captured nonlinear interactions between thermal and moisture

conditions. Soil GDD appeared most influential under transplanting, while air-based GDD was comparatively more important under direct seeding (Fig. 8). Building on cohort-based emergence tracking (at 10 % intervals), the phyllochron model reliably predicted leaf development stages, with strong correlations between observed leaf number and accumulated GDD (Fig. 9a), supporting its use in herbicide scheduling.

Model validation across multiple locations (2023–2024) revealed that most emergence and leaf development events clustered near zero day-differences between predictions and actual observations, underscoring the model's temporal accuracy. Some deviations occurred at later emergence phases or advanced leaf stages, likely reflecting site-

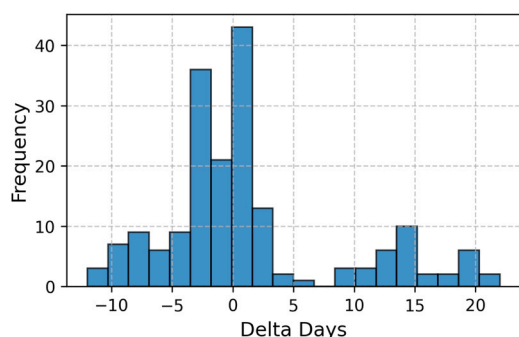


Fig. 11. Distribution of delta days, representing the difference in days between predicted and observed leaf emergence dates (year 2024). This result, obtained from the integrated RandomForest regression and phyllochron models, demonstrates the model's temporal accuracy; a narrow distribution with lower absolute delta_days indicates higher precision in predicting leaf development timing.

specific management practices or unmeasured environmental influences. Notably, soil-based GDD appeared to be the most influential predictor under transplanting, whereas air-based GDD was comparatively more important under both dry and wet direct seeding systems—consistent with the environmental drivers of each establishment method (Fig. 8). While these trends were consistent, differences in feature importance were not statistically tested and should be interpreted with caution. Overall, the integration of machine learning regressors with a mechanistic phyllochron model consistently delivered robust predictive performance and offers a promising framework to support weed management decisions, particularly for optimizing herbicide application timing across variable water regimes.

4.1. Model integration and performance

The coupling of a data-driven emergence model with a mechanistic phyllochron-based leaf development model provided a comprehensive framework to simulate Barnyardgrass phenology across different rice establishment methods. The Random Forest (RF) machine learning model accurately captured the complex, non-linear interactions governing weed emergence timing, while the phyllochron module translated post-emergence thermal time into leaf production. This integration leveraged the strengths of both approaches: RF handled heterogeneous environmental inputs (soil temperature, moisture, etc.) and their interactions without strong a priori assumptions, and the phyllochron model ensured that once emergence was predicted, subsequent leaf appearance followed biologically realistic thermal time progression. As a result, the combined model could forecast not only when barnyardgrass cohorts emerge but also how quickly they develop through leaf stages under varying field conditions. The hybrid modeling approach aligns with prior calls to use ecophysiological models in weed science to capture both emergence and growth processes for improved realism (Colucci et al., 2024; Gonzalez-Andujar et al., 2016; Karniadakis et al., 2021; Zhao et al., 2024). Numerous studies highlight the importance of environmental heterogeneity and biological drivers—such as dormancy release and germination—when predicting weed dynamics. For example, Maleki et al. (2023) and Reinhardt Piskackova et al. (2021) successfully applied thermal time models to forecast emergence in *Brassica napus* and *Amaranthus palmeri*, respectively, while Zandoná et al. (2024) validated TT/HTT models to predict *Ipomoea triloba* emergence. Methodological advances, including Onofri et al.'s (2022) unified time-to-event framework and soft computing methods reviewed by Marschner et al. (2024), underscore the value of integrating diverse data streams for more accurate predictions.

Recent advancements in agricultural modeling illustrate various

ways of blending ML with traditional agroecosystem and crop modeling techniques: Aderele et al. (2025) proposed ML meta-models to reduce computational costs in process-based models (PBMs); Li et al. (2023) merged APSIM with Random Forest to predict maize yield using climate data, NDVI, and extreme weather events; Xiao et al. (2022) showcased ML-based emulators that reproduce APSIM outcomes for yield and soil carbon dynamics; and Zhang et al. (2021) combined CERES-Maize with ML to assess climate-change impacts on maize hybrids. Furthermore, Sahoo et al. (2024) integrated UAV-based hyperspectral imaging with ML algorithms for precise estimation of wheat biophysical variables, and Zhao et al. (2024) coupled a Random Forest model with a novel change detection approach to forecast rice disease severity. Despite these advances, data scarcity (Aderele et al., 2025), regional specificity (Li et al., 2023), and interpretability (Z. Zhang et al., 2024) remain pressing concerns. Against this backdrop, our study builds on the consensus that multi-factorial inputs strengthen predictive accuracy (Colucci et al., 2024; Lundy et al., 2014). By incorporating both soil- and air-temperature GDD, cumulative precipitation, and crop establishment dates, we capture Barnyardgrass emergence patterns with greater nuance. We then segment predicted emergence into ten cohorts (each representing 10 % of total emergence) and apply a phyllochron model with fixed thermal thresholds (10°C base, 30°C ceiling) plus linear interpolation for fractional leaf development—an approach that addresses the nonlinear dynamics of emergence alongside the linearity of leaf appearance (Colucci et al., 2024; Gonzalez-Andujar et al., 2016).

4.2. Implications for weed management under climate variability

Predicting the timing of barnyardgrass emergence and development with greater precision has direct implications for field-level decision making, especially regarding herbicide applications and other control measures (Dionisi et al., 2023). The difference of a few days in weed emergence or reaching a particular leaf stage can determine the success of an intervention. By using our integrated model, practitioners can pinpoint the “control window” for barnyardgrass – for instance, applying a post-emergence herbicide when the majority of the weed cohort is at the 1–3 leaf stage (a stage known to be most susceptible to many herbicides) (Claerhout et al., 2015). For instance, model outputs suggest that 70 % of barnyardgrass cohorts reach the two-leaf stage by day of year (DOY) 150, indicating that applying a post-emergence herbicide between DOY 149 and 151 would likely maximize control efficacy. Barnyardgrass emerging within the first few weeks of the crop can drastically reduce rice yields and produce abundant seeds if not controlled (Awan and Chauhan, 2016); thus, aligning management actions with the model-predicted peak emergence or key growth stages is critical.

Accurate prediction of both weed emergence and leaf development stages underpins timely herbicide application and supports broader integrated management tactics. Several studies have stressed the importance of predicting key phenological windows for control. For instance, Reinhardt Piskackova et al. (2021) pinpointed 10 %, 50 %, and 90 % emergence thresholds in *A. palmeri*, facilitating targeted interventions. Brown et al. (2022) extended such insights to multiple weed species, underscoring that peak emergence timing significantly differs among early-, mid-, and late-season weeds. Our findings, showing the primacy of soil-based GDD under transplanting systems and the significance of air-based GDD for direct-seeded systems, highlight the necessity for tailored strategies, especially in environments where establishment methods and climate regimes vary.

Moreover, the integrated framework aligns with the broader goal of reducing herbicide reliance, which remains a central challenge in contemporary weed science (Batlla and Luis Benech-Arnold, 2007; Ip et al., 2018). By precisely forecasting emergence and leaf-stage progression, our model can help identify narrower windows for effective herbicide application, echoing the precision-focused approaches of Nikolić et al. (2021) combined UAV-based weed detection with

predictive models to cut herbicide use by up to 93 %. In addition, informed timing could mitigate the rapid evolution of herbicide resistance, as managers can rotate or mix herbicides more effectively when they know the most vulnerable weed growth stages (Gonzalez-Andujar et al., 2016). Thus, beyond the immediate benefit of optimized control, an integrated emergence and leaf development prediction model contributes to a more sustainable, data-driven approach to weed management.

Finally, these strategies gain importance under increasing climate variability. With erratic weather patterns, weed emergence is becoming less predictable by static calendars. A model-driven approach ensures that herbicide and cultivation timings dynamically adjust to each season's conditions, accounting for warmer or cooler spells that accelerate or delay weed growth. For instance, a hotter-than-average spring could lead to barnyardgrass emerging in higher numbers and much earlier; if farmers rely on historical averages, they might miss the optimal control window, whereas a GDD-based prediction would capture this shift. Likewise, unexpected cold snaps or drought could pause weed emergence, and the model would prevent unnecessary early herbicide sprays, preserving resources for when weeds truly emerge. By improving timing precision, our model can increase herbicide efficacy, reduce excessive chemical use (by targeting weeds at their most sensitive stages), and ultimately mitigate yield losses. This precision is especially valuable as climate change affects weed biology – research suggests that weed germination and emergence patterns will shift with changing temperature and moisture regimes (Boddy et al., 2012). Incorporating real-time weather data and climate-adjusted parameters into decision-making tools will help sustain effective weed control even as conditions deviate from the historical norm (Lundy et al., 2014). In summary, the integration of our emergence and development model into weed management protocols enables proactive, well-timed interventions that are resilient to climate variability, thereby improving control outcomes and potentially reducing the selection pressure for herbicide resistance due to fewer missed or late treatments.

4.3. Limitations and future directions

Despite demonstrating robust predictive performance in standard field conditions, our integrated model revealed discrepancies during peak emergence flushes and later leaf stages—a finding that aligns with broader literature on the unpredictability of real-world scenarios (Colucci et al., 2024; Onofri et al., 2022). For instance, Maleki et al. (2023) underscore how germination synchrony can yield concentrated emergence events once critical physiological thresholds are surpassed, potentially challenging simpler model assumptions. Data quality also remains a pivotal concern; accurate, site-specific soil temperature measurements are especially crucial in transplanting systems, where relying solely on air temperature proxies could compromise prediction accuracy. Additionally, although the current model focuses on thermal time and moisture inputs, it does not account for several other influential environmental and management factors. For instance, variability in soil nutrient levels (e.g., nitrogen, phosphorus, and potassium) can significantly affect seedling vigor and the rate of leaf development. Differences in light availability, such as shading from adjacent crops, may slow phyllochron progression, while tillage history—including the depth and timing of soil disturbance—can influence seed dormancy status and alter seedbank distribution. These factors were not explicitly parameterized in the current framework. Incorporating such variables, along with realistic stress functions to capture shading or herbicide effects, could help explain additional variance and improve model robustness. Furthermore, since core parameters such as base temperature and phyllochron slope were derived from Japanese barnyardgrass biotypes, multi-year and multi-site trials are essential to evaluate the model's transferability. Application to other regions may require recalibration to reflect differences in local climates and weed ecotypes. Although integrating machine learning with phyllochron modeling

enhances predictive precision, its reliance on timely, high-resolution data—such as real-time soil temperature—may constrain widespread on-farm adoption without adequate sensor infrastructure. Lastly, one limitation is that the emergence and leaf development models were trained on data from 2023, which did not capture extreme weather events such as heatwaves or flooding; thus, predictions under such conditions may involve extrapolation and reduced reliability.

Beyond data and ecological factors, practical adoption poses another challenge. Integrating remote sensing technologies—like UAV-based monitoring (Nikolić et al., 2021) and real-time temperature feeds (Lundy et al., 2014), could help bridge this gap by improving spatial and temporal resolution. Additionally, incorporating advanced analytics (e.g., big data approaches, Markov models) may enable integration of economic and operational considerations into weed management strategies. To ensure practical utility—whether for herbicide scheduling, crop sowing, or irrigation planning—collaboration with extension services to develop accessible user interfaces and farmer training modules will be critical for translating model outputs into actionable, field-ready decisions.

5. Conclusion

Notwithstanding the current constraints, our integrated approach offers a foundation for more adaptive and predictive weed management, especially as climate variability challenges historical assumptions about weed emergence. Future refinements—encompassing expanded validation efforts, incorporation of additional environmental variables, and improved end-user accessibility—promise to bolster both accuracy and utility. Ultimately, merging machine learning with process-based crop and weed models, as demonstrated here, can serve as a blueprint for managing other problematic weed–crop systems and contribute to the broader goal of creating resilient and resource-efficient agroecosystems in a rapidly changing world.

CRedit authorship contribution statement

Holger Hoffmann: Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition. **Kromminga Heiko:** Writing – review & editing, Software, Data curation. **Akhter Muhammad:** Writing – review & editing, Validation, Methodology, Data curation, Conceptualization. **Gang Zhao:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.eja.2025.127846](https://doi.org/10.1016/j.eja.2025.127846).

Data availability

Data will be made available upon request. Please note that pro-

proprietary data used in this study is not available due to commercial restrictions.

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